

# Applied Data Science Capstone Project Report

SpaceX Falcon 9 First-Stage Landing Prediction & Economic Analysis

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## 1. Executive Summary

In the competitive commercial spaceflight industry, understanding the economic feasibility of reusable launch systems is paramount. SpaceX successfully undercuts traditional aerospace competitors by recycling the first stage of its Falcon 9 launch vehicle. The primary objective of this capstone project is to predict whether a Falcon 9 first stage will successfully land (either on an autonomous droneship or ground landing pad) based on operational factors such as payload mass, orbit type, launch site, and flight history.

Using a comprehensive data science pipeline—spanning REST API ingestion, web scraping, data wrangling, SQL querying, advanced Exploratory Data Analysis (EDA), interactive Folium mapping, Plotly Dash dashboards, and predictive classification modeling—we evaluated historical mission performance using the code and notebooks available in the public project repository: <https://github.com/eortiz26/applied-data-science-capstone>.

### Key Findings & Results

- **Launch Site Performance:** KSC LC-39A demonstrates the highest overall success rate for booster recovery, benefiting from ideal geographic and logistical orientation.
- **Payload Dynamics:** Lighter payloads correlate with significantly higher recovery success rates due to fuel reserve margins required for boost-back and entry burns.
- **Predictive Accuracy:** Machine learning algorithms (specifically Support Vector Machines and Logistic Regression) successfully predict landing outcomes with high out-of-sample accuracy, offering actionable intelligence for launch cost modeling.

## 2. Introduction & Project Background

SpaceX advertises Falcon 9 launches at approximately \$62 million—substantially cheaper than competing providers who discard their launch vehicles. The ability to land and reuse the first booster stage is the core driver of this cost advantage. Predicting whether a launch will achieve a controlled landing allows investors, insurers, and competitor analysts to gauge mission risk and operational efficiency. All supporting Python scripts and Jupyter notebooks can be inspected directly at <https://github.com/eortiz26/applied-data-science-capstone>.

## 3. Methodology Workflow

Our methodology adhered to a structured data science framework, moving sequentially from raw data acquisition to predictive deployment:

### Phase A: Data Collection

**SpaceX REST API:** Queried `api.spacexdata.com/v4/` to extract comprehensive launch logs, core details, payloads, and launchpads.

**Web Scraping:** Extracted historical launch tables from Wikipedia utilizing BeautifulSoup to cross-reference records.

### Phase B: Data Wrangling & EDA

**Wrangling:** Handled missing values, filtered dataset attributes, and engineered a binary target variable (Class: 1 for successful landing, 0 for failure).

**EDA:** Performed bivariate visualization via Matplotlib/Seaborn and relational querying via SQLite.

### Phase C: Interactive Analytics

**Folium Mapping:** Built interactive geospatial maps examining launch site proximities to coastlines, highways, and railways.

**Plotly Dash:** Developed a dynamic web application featuring interactive success pie charts and payload range sliders.

### Phase D: Predictive Modeling

**Classification:** Trained Logistic Regression, SVM, Decision Trees, and KNN models with hyperparameter tuning via GridSearchCV.

## 4. Results & Exploratory Analysis

Exploratory data analysis uncovered distinct operational patterns associated with successful booster recovery. The complete code and generated assets are cataloged in our public repository: <https://github.com/eortiz26/applied-data-science-capstone>.

### SQL Query Highlights & Insights

| Analysis Objective           | Primary Finding / Output Summary                                                                                    |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Unique Launch Sites          | Identified active sites: CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E.                                                 |
| NASA Payload Contribution    | Calculated total payload mass carried for NASA CRS missions, highlighting institutional payload distributions.      |
| Booster F9 v1.1 Analysis     | Computed average payload mass for early Falcon 9 v1.1 iterations, noting structural capacity constraints.           |
| Landing Outcomes (2010–2017) | Ranked landing outcomes between June 4, 2010 and March 20, 2017, demonstrating steep technological learning curves. |

### Interactive Dashboard & Geospatial Findings

**Folium Proximity Analysis:** Mapping launch sites relative to geographic features revealed that sites are strategically located on eastern coastlines (e.g., Cape Canaveral) to maximize Earth's rotational velocity during eastward launches while maintaining safe ocean recovery corridors away from dense population centers.

**Plotly Dash Insights:** The interactive payload slider demonstrated that payloads below 5,000 kg achieve significantly higher landing success rates on droneships and ground pads compared to heavy commercial payloads exceeding 9,000 kg, which consume primary fuel reserves during ascent.

## 5. Predictive Analysis & Model Evaluation

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We trained four distinct machine learning classifiers (Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors) on our preprocessed feature set after applying Standard Scaler normalization and train-test splitting.

### Classification Performance Summary

- **Hyperparameter Tuning:** GridSearchCV successfully identified optimal regularization parameters ( $C$ ,  $\lambda$ ), penalty types and neighbor counts ( $k$ ), preventing overfitting across models.
- **Model Accuracy:** Logistic Regression and SVM achieved top out-of-sample prediction accuracies (~83%), effectively distinguishing between successful and failed landing trajectories.
- **Confusion Matrix Evaluation:** The confusion matrix analysis indicated low false-negative rates, confirming that the models reliably flag high-risk launches where booster recovery is unlikely.

## 6. Conclusions & Future Recommendations

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This capstone project successfully demonstrated that machine learning algorithms can predict Falcon 9 first-stage landing outcomes with high fidelity based on historical telemetry, payload metrics, and orbital destinations.

**Strategic Takeaways:** Launch providers can utilize these predictive models to dynamically adjust mission profiles, optimize fuel allocation for heavier payloads, and refine economic forecasting for reusable rocket fleets.

**GitHub Repository Reference:** All completed notebooks, Python scripts, API ingestion scripts, and Plotly Dash applications are fully documented and accessible for peer review at <https://github.com/eortiz26/applied-data-science-capstone>.